

Stigix Private & Hybrid Deployment Topologies Guide

This guide provides a comprehensive, pedagogical overview of all deployment scenarios supported by Stigix. It explains how the **Control Plane** (Fleet Gateway WebSocket Tunnels, Local Registry, and Global Provisioning) and the **Data Plane** (Traffic Generator, DEM, VoIP, IPerf, and Security) operate across private labs, enterprise LAN/WANs, and multi-cloud environments (AWS, GCP, Azure, Hetzner) **without requiring inbound firewall ports or public IP addresses on the Leader.**

Architecture Overview & Key Concepts

Stigix distinguishes between two communication layers:

STIGIX COMMUNICATION LAYERS
1. CONTROL PLANE (Layer 7 – WebSocket & HTTP RPC) - Fleet Gateway Tunnels (M5 Inbound & M6 Outbound Reverse I - Telemetry streaming (CPU, RAM, interface status every 15s - Single Pane of Glass Proxy UI (1-click remote node contr - Global Provisioning Sync (Apps, Custom TCP, Probes, SLA,
2. DATA PLANE (Layer 3/4 – IP Synthetic & Security Traffic) - IPerf3 throughput tests & Bandwidth Stress - Real-Time VoIP SIP/RTP audio streams - Synthetic DEM SaaS probes (Office 365, Salesforce, AWS) - EICAR & Security malware inspection flows - Custom TCP Application daemons (SAP, POS, HL7, Banking)

The 4 Deployment Topologies

```
flowchart TB
    subgraph S1["Scenario 1: On-Premise LAN / MPLS"]
        L1["Leader (DC1: 192.168.1.10)"] ---|"Direct HTTP / LAN"|
    end

    subgraph S2["Scenario 2: Remote Branch behind NAT/CGNAT"]
        P2["Spoke behind NAT (Home / 4G)"] -->|"M5: Spoke dials I"|
    end

    subgraph S3["Scenario 3: Private Leader + Cloud Peers (Zero I"]
        L3["Leader (Private RFC1918 LAN - 0 Inbound Ports)"] -->|
    end

    subgraph S4["Scenario 4: Public SaaS Leader"]
        L4["Leader (Public FQDN / Cloudflare)"] <-->|"Cloudflare"|
    end
```

Scenario 1: Private Leader + Private Spokes (Standard On-Premise / LAN / MPLS)

Use Case

You are running Stigix inside an enterprise data center or private lab where all nodes reside on the same Layer 2/3 network (e.g. 192.168.122.0/24, 192.168.123.0/24) or are interconnected via MPLS/corporate VPN.

```
sequenceDiagram
    autonumber
    participant DC1 as DC1 Leader (192.168.122.51)
    participant BR1 as BR1 Branch Peer (192.168.122.57)
```

```
Note over BR1,DC1: Control Plane (Direct HTTP)
BR1->>DC1: HTTP POST /api/registry/heartbeat (every 15s)
DC1-->>BR1: 200 OK (Peer Registered)
BR1->>DC1: HTTP GET /api/provisioning/pull
DC1-->>BR1: Returns configuration bundles
```

```
Note over BR1,DC1: Data Plane (Direct IP Routing)
BR1->>DC1: IPerf3 / Voice RTP / TCP Custom Traffic
```

- **Control Plane:** Spokes send direct HTTP heartbeats to `http://<LEADER_IP>:8080/api/registry`.
 - **Data Plane:** Full bidirectional routing between all nodes.
 - **Configuration:**
 - **Leader:** Select Role: Forced Leader or set `STIGIX_REGISTRY_MODE=leader`.
 - **Spoke:** In Settings -> Target Controller, set *Leader IP / FQDN* to `http://192.168.122.51:8080/api/registry`.
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Scenario 2: Private Leader + Remote Spokes behind NAT / CGNAT (Inbound Tunnel M5)

Use Case

The Leader resides in a central data center (with port 8080 reachable from branches), while branch nodes are deployed in home offices, retail shops, or LTE/5G routers behind strict NAT or CGNAT where **no inbound ports can be opened on the branch**.

```
sequenceDiagram
    autonumber
    participant DC1 as DC1 Leader (Hub)
    participant Spoke as Spoke (Behind Home NAT / 4G)

    Note over Spoke,DC1: M5: Spoke initiates outbound WebSocket c
    Spoke->>DC1: WS Connect to ws://<DC1_IP>:8080/fleet-tunnel
    DC1-->>Spoke: WS Established (Persistent L7 Tunnel)

    loop Every 15s
```

```
Spoke->>DC1: WS peer:telemetry (CPU, RAM, Status)
end
```

```
Note over DC1,Spoke: Operator clicks Spoke in DC1 UI (Zero Po
DC1->>Spoke: WS gateway:forward (GET /api/status)
```

```
Spoke-->>DC1: WS gateway:forward response
```

- **Control Plane:** Spoke opens an outbound WebSocket connection to the Leader. Once open, the Leader can proxy UI requests and inspect logs in real time.
- **Data Plane:** Spoke can generate traffic toward the Leader; Leader cannot originate direct IP traffic into the Spoke unless routed through an SD-WAN overlay tunnel.
- **Configuration:**

- **Spoke:** In Settings -> Target Controller, configure the Leader URL. The WebSocket client connects automatically.
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Scenario 3: Private Leader + Multi-Cloud Peers (Outbound Reverse Dial M6 — Zero Inbound on Leader)

Use Case

This is the most powerful and secure hybrid deployment.

- Your Leader resides in your **Private Home LAN / Lab** (e.g. 192.168.122.51) behind a standard internet router. **You do NOT want to open any port on your firewall or expose your lab to the Internet.**
- You deploy Stigix nodes on Public Cloud VPS providers (**Hetzner, AWS EC2, GCP Compute Engine, Azure VM**).

```
sequenceDiagram
    autonumber
    actor Admin as Admin (Lab UI)
    participant DC1 as DC1 Private Leader (LAN)
    participant Cloud as Hetzner / AWS Peer (Public IP: 142.132.1

    Admin->>DC1: Adds Manual Target "Hetzner" (142.132.193.157:80
```

```

Note over DC1,Cloud: M6: Leader initiates outbound connection
DC1->>Cloud: WS Connect to ws://142.132.193.157:8080/fleet-tu
Cloud-->>DC1: WS Tunnel Established (Persistent L7 Link)

Note over Cloud,DC1: Automated Provisioning Sync over WS
Cloud->>DC1: WS provisioning:get_manifest
DC1-->>Cloud: Manifest (Revisions & Checksums)
Cloud->>DC1: WS provisioning:pull_bundle (custom-tcp-apps, ap
DC1-->>Cloud: Bundle Payloads
Cloud->>Cloud: Hot-reloads TCP Servers & SaaS Probes (rev 1 a

loop Telemetry Stream
    Cloud->>DC1: WS peer:telemetry (every 15s)
    DC1->>DC1: Ingests into Local Registry (Status: Online <
end

Note over Admin,Cloud: Single Pane of Glass Control
Admin->>DC1: Switches to "⚡ Hetzner" in top bar
DC1->>Cloud: WS gateway:forward (Executes tests on Cloud)
Cloud-->>DC1: Returns results to Leader UI

```

Why this works without configuration on the Cloud Peer:

1. **Zero-Touch Cloud Setup:** Start the container on the VPS (`docker run -d -p 8080:8080 jsuzanne/stigix:v2`). No need to log into the VPS or configure Leader IP.
2. **Leader Adoption:** Add the target host in DC1 -> Settings -> Stigix Targets. DC1 immediately dials out to the VPS.
3. **Bi-Directional L7 Control:**
 - Hetzner streams real-time telemetry back to DC1.
 - Hetzner pulls all configuration bundles (`custom-tcp-apps`, SaaS apps, connectivity probes) over the WebSocket tunnel.
 - Hetzner status switches to ⚡ **WS Tunnel Synced** with all green revision badges.
4. **Data Plane Behavior:**
 - Hetzner acts as an external reflector / responder for traffic originating from branches (`BR1 -> Hetzner`).
 - Hetzner can execute synthetic tests toward public SaaS/Cloud endpoints (Office 365, AWS, Salesforce) controlled directly from DC1's

Scenario 4: Public SaaS Leader + Distributed Global Mesh

Use Case

You deploy the Stigix Leader on a public cloud instance with a valid DNS name (e.g. `https://stigix.company.com`) or via Cloudflare Worker Registry.

- **Control Plane:** All spoke nodes globally register to the central registry.
- **Data Plane:** Full mesh testing across multi-region networks.
- **Configuration:** Nodes use Cloudflare Worker Autodiscovery (`STIGIX_REGISTRY_URL`) or direct HTTPS URLs.

Feature & Protocol Compatibility Matrix

Capability	Scenario 1 (LAN/MPLS)	Scenario 2 (NAT Spoke M5)	Scenario 3 (Cloud Peer M6)	Scenario 4 (Public SaaS)
Inbound Ports required on Leader	None (LAN only)	Port 8080	0 (Zero)	Port 443 / 8080
Inbound Ports required on Peer	None (LAN only)	0 (Zero)	Port 8080	0 (Zero)
Telemetry Streaming (CPU/RAM/Health)	✓ HTTP Poll (15s)	✓ WS Push (15s)	✓ WS Push (15s)	✓ HTTP / WS

Single-Pane-of-Glass Remote Proxy UI	✓ Direct	✓ WS Proxy	✓ WS Proxy	✓ Direct / WS
Global Provisioning Sync	✓ HTTP Pull	✓ WS Pull/Push	✓ WS Pull/Push	✓ HTTP Pull
Custom TCP Apps Server Mode	✓	✓	✓	✓
Traffic Generation from Branch to Node	✓	✓	✓	✓
Traffic Generation from Node to Branch	✓	Requires VPN	Requires VPN	Requires VPN

🔧 Step-by-Step Runbook: Adding a Cloud VPS Target (Scenario 3)

Step 1: Deploy Stigix on Cloud VPS (e.g. Hetzner, AWS, GCP, Azure)

Run standard Docker on the remote server:

```
docker run -d \
  --name stigix \
  --restart unless-stopped \
  -p 8080:8080 \
  jsuzanne/stigix:v2
```

Step 2: Add Target on your Private Leader (DC1)

1. Open your private Leader dashboard (<https://sdwandc1.carenaje.fr> or <http://192.168.122.51:8080>).

2. Navigate to **Settings** → **Stigix Targets**.

3. Click + **ADD TARGET**:

- **Target Name:** Hetzner-Cloud
- **Host / IP:** 142.132.193.157
- **Port:** 8080
- **Capabilities:** Check Voice, Speedtest, Security, Failover.

4. Click **Save**.

Step 3: Verify Automated Adoption & Sync

1. Navigate to **Settings** → **Target Controller**:

- The Cloud node will show status  **WS Tunnel Synced**.
- Clicking **TEST** verifies the WebSocket link and displays round-trip latency (RTT: ~18ms).

2. Navigate to **Mesh Overview**:

- Hetzner-Cloud displays with the  **WS Tunnel** badge.

3. Use the **Instance Switcher** in the top navigation bar to select  Hetzner-Cloud and control the remote node directly from your private lab.

Revision History

Date	Stigix Version	Author / Trigger	Summary of Changes
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2026-09-30	v2.0.111	Stigix Core Team	Initial creation of Private & Hybrid Deployment Topologies Guide covering M5/M6 WebSocket Tunnels, Zero- Inbound Leader, and Multi- Cloud Provisioning Sync
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